Within-host malaria parasite clonal dynamics impact on dry-season asymptomatic persistence

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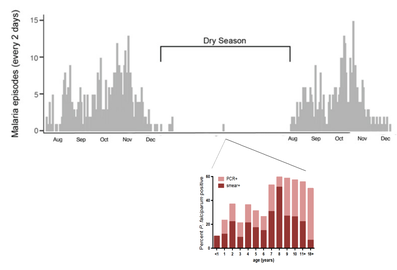
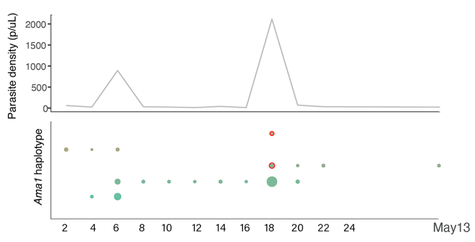
In Mali malaria cases occur in the wet season



During the dry season parasites persist in a fraction of asymptomatic hosts

But not all parasites bridge two wet seasons

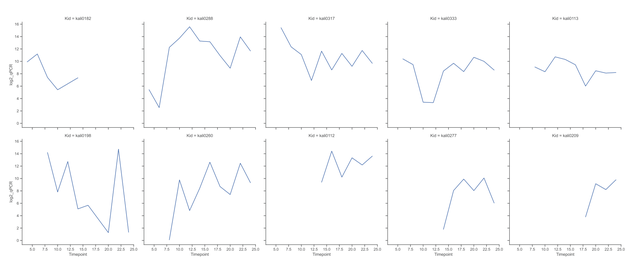


Ama1 gene clones appear and disappear rapidly throughout the transmission season

Longitudinal cohort study following ~600 individuals

129 children

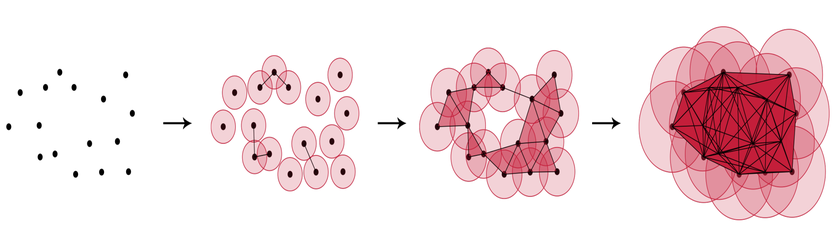
92 PCR+ in May, 2013
37 PCR- in May, 2013



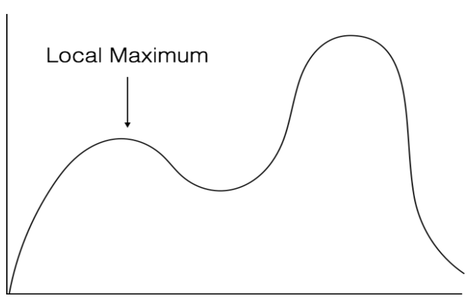
12 biweekly backwards timepoints from July-Dec

We need a formal method to characterise peaks of parasite density for each individual

Methods



Persistent homology



Local maxima

References
Andrade, Carolina M. et al. (2024) EMBO Molecular Medicine.
Andrade, Carolina M. et al. (2020) Nature Medicine.
Edelsbrunner, H. & Harer, J. (2010) American Mathematical Soc.
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Summary

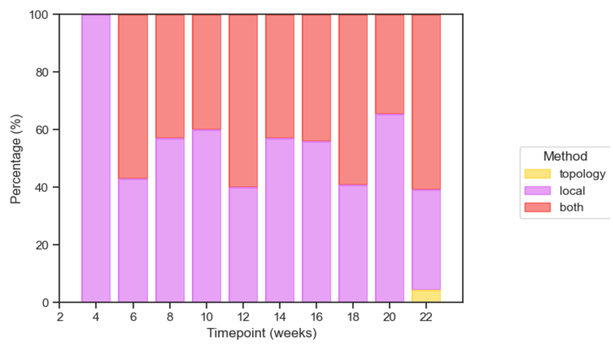
- High clonal turnover of malaria parasites during transmission season
- Not all clones bridge two wet seasons
- Need a method for identifying peaks in noisy and dinamic parasite density data

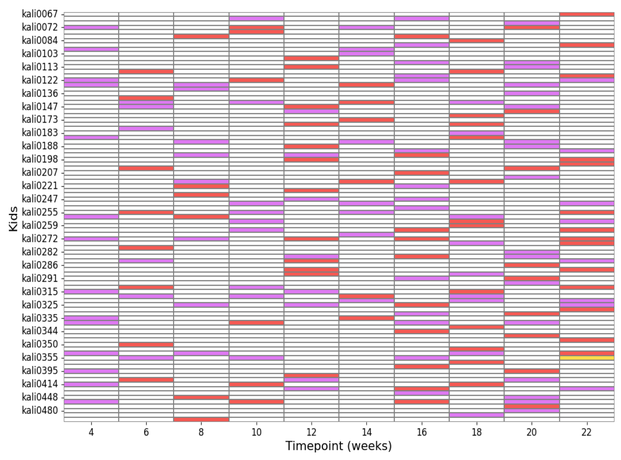
Key questions

What are the main differences between parasites of persistent and resolved infections?
Are peaks of parasite density linked with appearance and disappearance of clones?
Does this culminate in persistence or clearance of infections?

Clonal dynamics over time in a person affects dry-season persistence

Results





In average

Local maxima identifies 2 ± 1.7 peaks per kid and Persistent homology 0.92 ± 0.6 peaks per kid

Future work

- Count number of first apperances per peak in persistent and cleared infections
- Assess the accuracy and specificity of peak detection by both methods